

◆ **1. Project Title**

Server-Client Network Setup using Wireless Router with Domain, DNS, and Resource Sharing

◆ **2. Objective**

To design and implement a small-scale network infrastructure using a wireless router, where a Windows Server acts as a Domain Controller providing services like DNS, HTTP, user management, and file sharing to connected clients.

◆ **3. Network Topology**

- **Server (Windows Server 2022)**
 - IP Address: 192.168.1.10
 - Role: Domain Controller, DNS Server, File Server, Web Server
 - **Client Machine (Windows 10/11)**
 - IP Address: 192.168.1.11
 - Role: Domain User System
 - **Wireless Router**
 - Used for communication between server and client
 - No internet required (LAN-based setup)
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◆ **4. Requirements**

Hardware:

- 1 Physical Server Machine
- 1 Client Machine
- 1 Wireless Router

Software:

- Windows Server 2022
 - Windows 10/11 (Client)
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◆ 5. Configuration Steps

Step 1: Network Configuration

- Assigned static IP to server: 192.168.1.10
 - Assigned static IP to client: 192.168.1.11
 - Subnet Mask: 255.255.255.0
 - DNS set to server IP (192.168.1.10)
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Step 2: Install Active Directory Domain Services (AD DS)

- Installed AD DS role on server
 - Promoted server to Domain Controller
 - Created new domain (example: lab.local)
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Step 3: User and Group Management

- Created Organizational Units (OU)
 - Created users inside OU
 - Created security groups
 - Added users to groups
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Step 4: Folder Sharing & Permissions

- Created shared folders
 - Assigned permissions based on groups
 - Verified access from client system
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Step 5: DNS Configuration

- Configured DNS during AD setup
 - Verified name resolution using ping and nslookup
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Step 6: HTTP (Web Server Setup)

- Installed IIS (Internet Information Services)
 - Hosted a basic web page
 - Accessed website from client browser using server IP/domain
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Step 7: Client Domain Join

- Connected client to domain (lab.local)
 - Logged in using domain user credentials
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6. Testing & Validation

-  Ping test between server and client
 -  Domain login successful
 -  Shared folder access verified
 -  DNS resolution working
 -  Web server accessible from client
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7. Outcome

Successfully implemented a working client-server network with:

- Domain Controller
 - DNS Server
 - File Sharing System
 - Web Server
 - User & Group Management
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◆ 8. Learning Outcome

- Practical understanding of Windows Server roles
 - Hands-on experience with Active Directory
 - Knowledge of network configuration and troubleshooting
 - Experience in real-time enterprise-like environment
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◆ 9. Future Improvements

- Add DHCP Server
- Implement Group Policy (GPO)
- Add multiple clients
- Configure firewall and security policies

